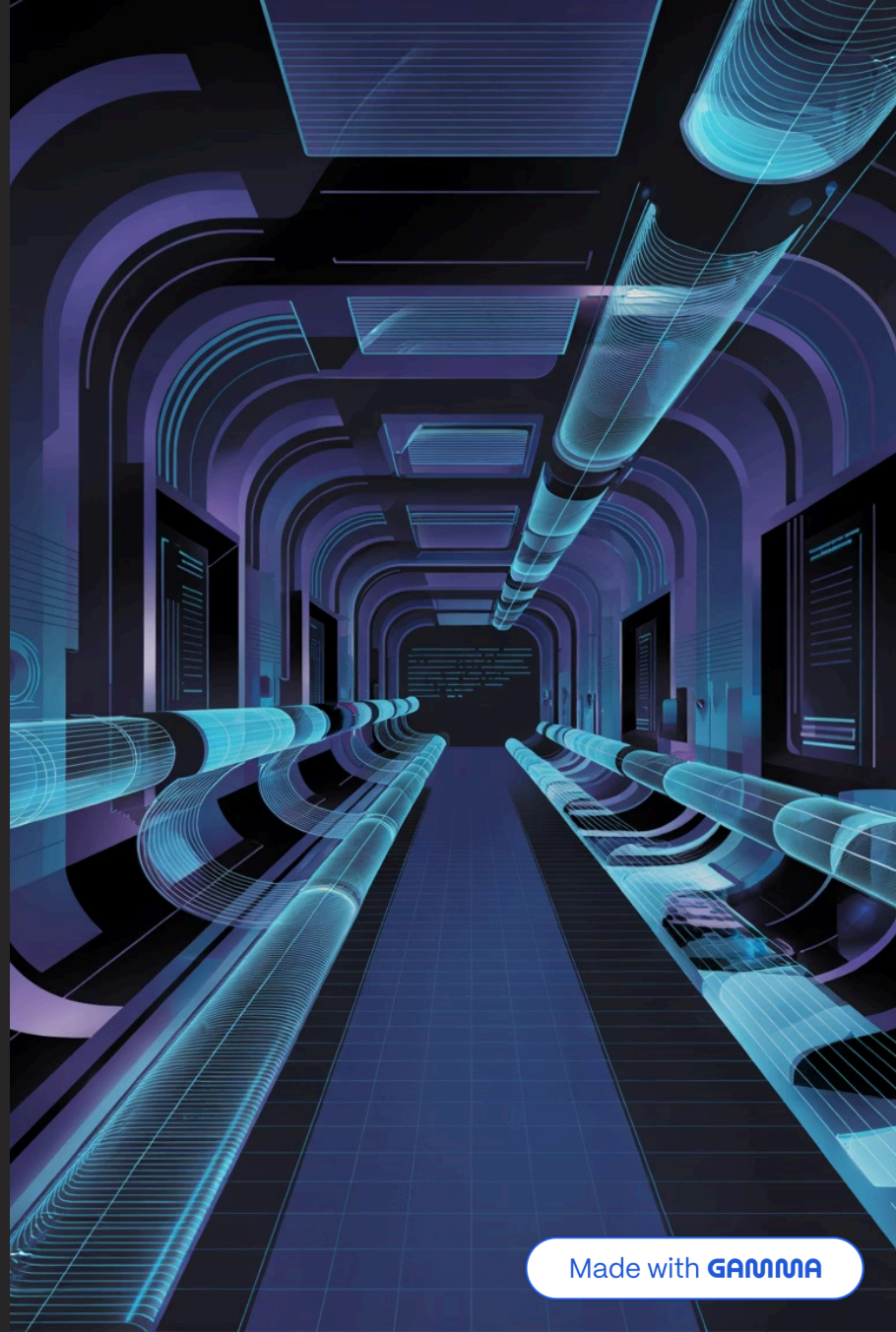
Virtual Private Network Architecture

Secure, encrypted communication across public networks

By Harsh Pandey – PRN: 2414110071



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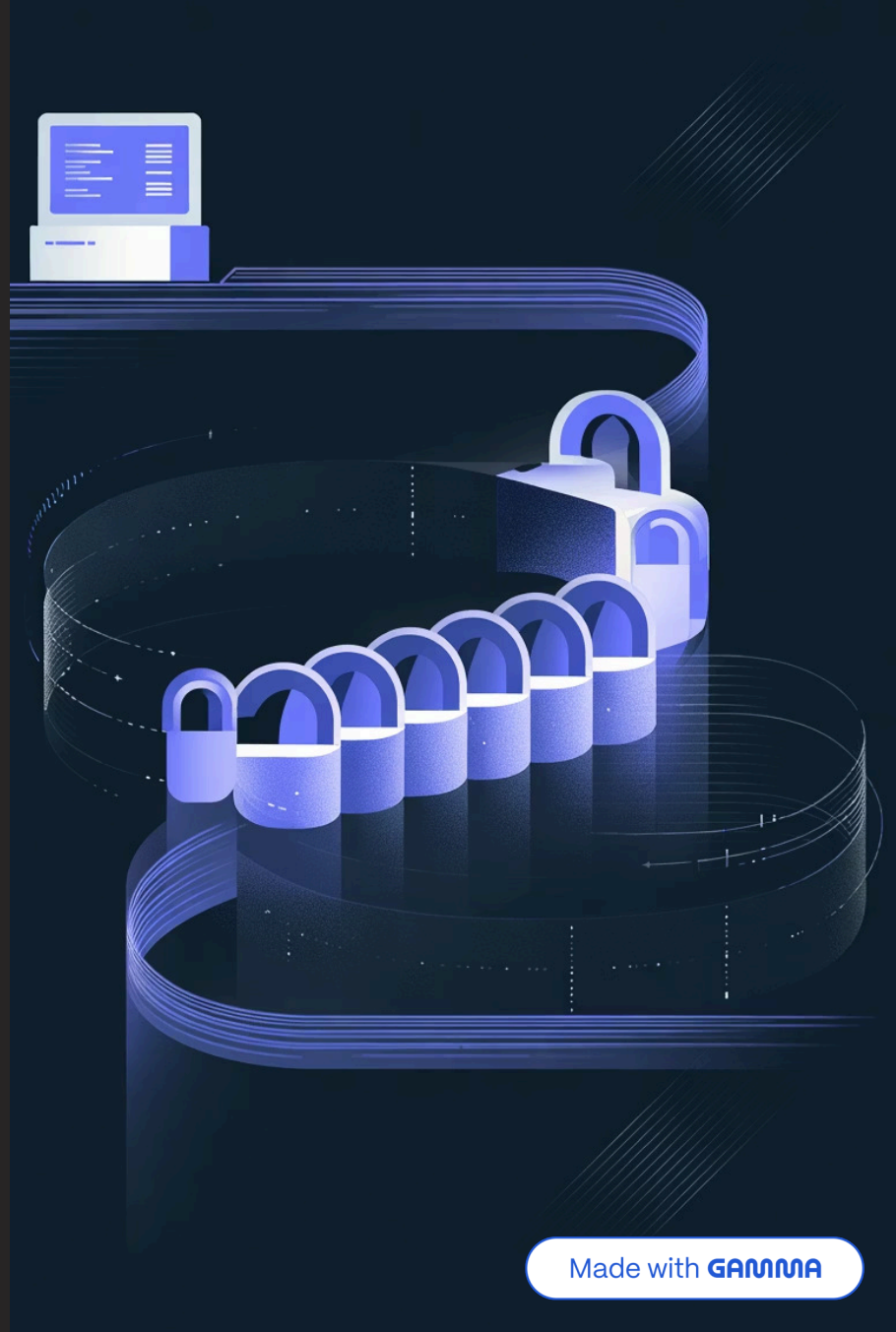
What is a VPN?

Core Definition

A secure, encrypted tunnel that transmits data across public networks while maintaining privacy and confidentiality.

Key Components

- VPN Client
- VPN Gateway
- Encryption Protocols
- Tunneling Protocols



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Core Objectives

Data Protection

Safeguard confidentiality and integrity of sensitive information.

Remote Access

Enable secure connectivity for distributed workforces.

Authentication

Verify user identity and control access permissions.

Cost Efficiency

Deliver scalable, affordable communication infrastructure.

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Key Advantages

Strong Encryption

Military-grade encryption ensures data privacy and security.

Mobile Workforce Support

Enables secure access from anywhere, supporting remote and hybrid work.

Reduced Costs

Eliminates expensive WAN infrastructure while maintaining performance.

Cloud Integration

Seamlessly connects with cloud and hybrid network environments.

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Important Limitations

Performance Issues

Encryption overhead can introduce latency and reduce throughput.

Security Risks

Potential for misuse or hidden attacks if not properly managed.

Configuration Complexity

Setup and maintenance require specialized expertise and resources.

Single Point of Failure

Gateway can become a critical vulnerability if compromised.

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Real-World Applications

Remote Access

Secure connectivity for employees working outside the office.

Branch Connectivity

Encrypted links between multiple office locations.

Public Wi-Fi Protection

Shields users from threats on unsecured networks.

Cloud Integration

Secure access to cloud services and hybrid infrastructure.

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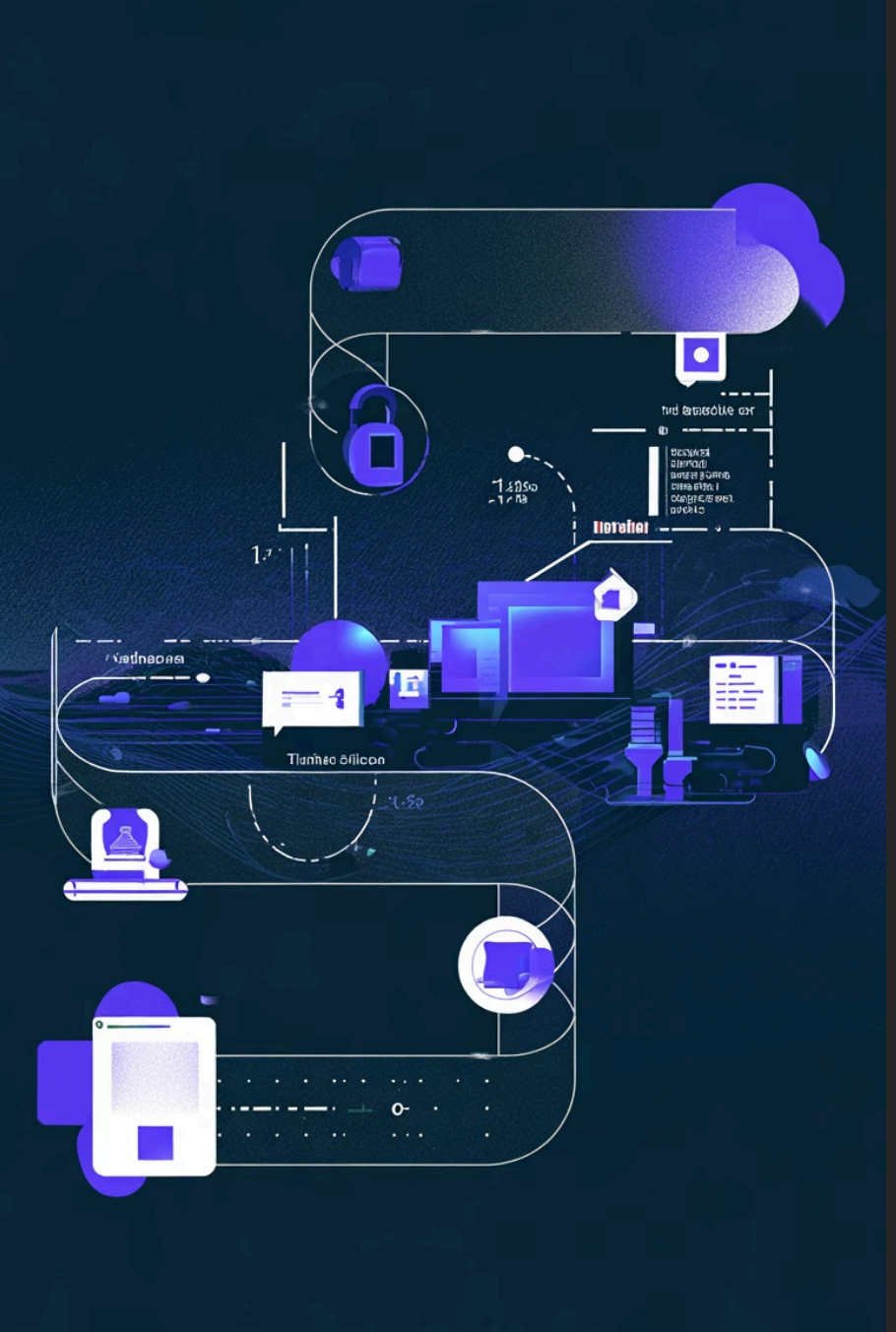

Modern VPN Evolution

1. **Traditional VPN**

   Perimeter-based security model.

2. **Zero Trust Network Access (ZTNA)**

   Verify every access request, never trust by default.

3. **SD-WAN Integration**

   Software-defined networking for enhanced flexibility and control.

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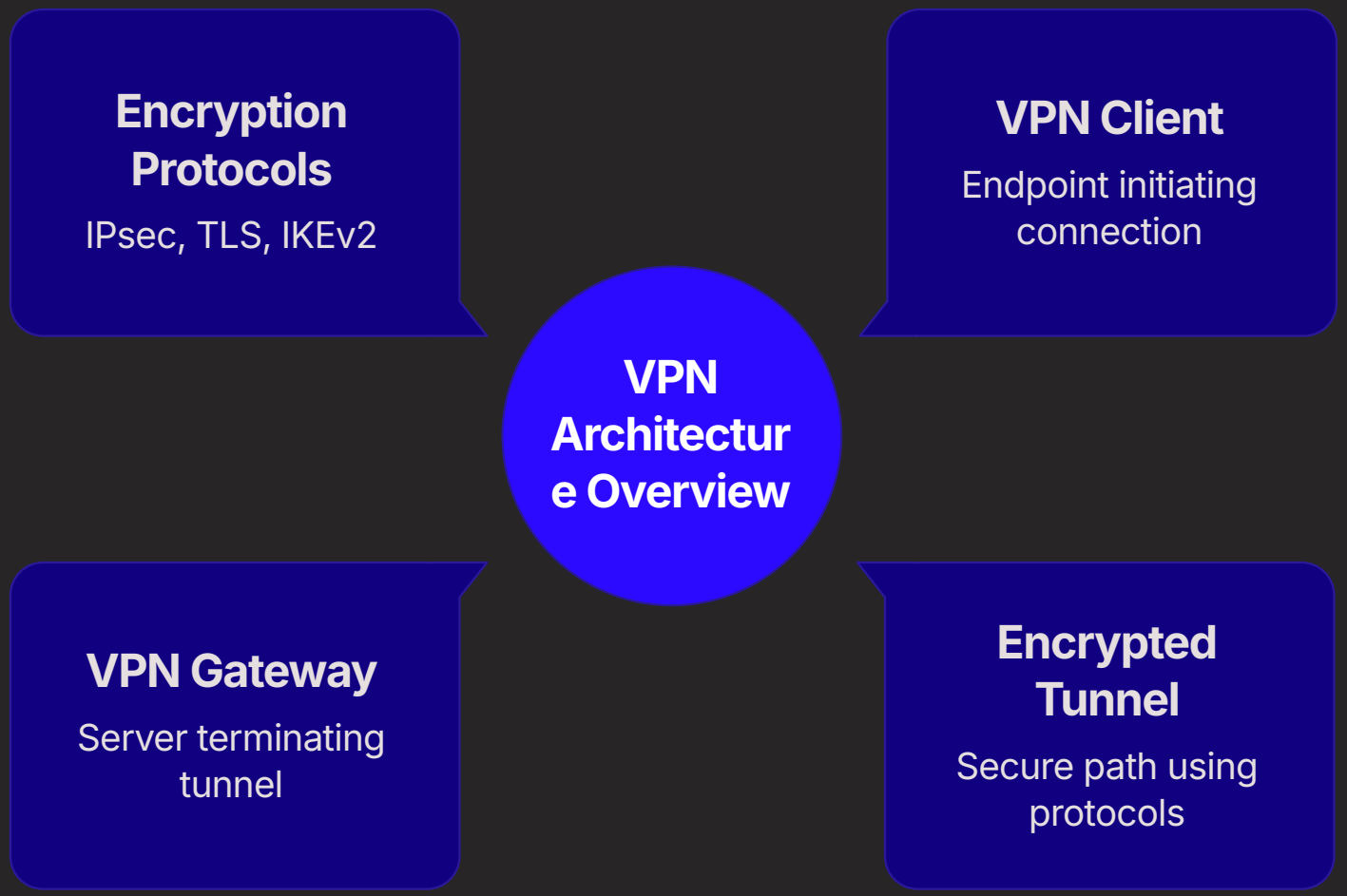
VPN Architecture Overview



The architecture demonstrates how client devices establish secure connections through encrypted tunnels to gateway servers, protecting data in transit.

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Why VPNs Matter Today

Remote Work Essential

Enables secure access for distributed teams across the globe.

Cloud Security

Protects data moving between on-premises and cloud environments.

Privacy Protection

Ensures confidential communication in an increasingly connected world.

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VPNs: The Foundation of Secure Communication

1. **Ensure secure and private data exchange**

   VPNs protect sensitive information across public networks.

2. **Essential for modern work environments**

   Support remote access, cloud integration, and hybrid infrastructure.

3. **Evolving with modern security standards**

   Integrate ZTNA and SD-WAN for enhanced flexibility and protection.

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